

CHAPTER 12

The Nervous System and Nervous Tissue

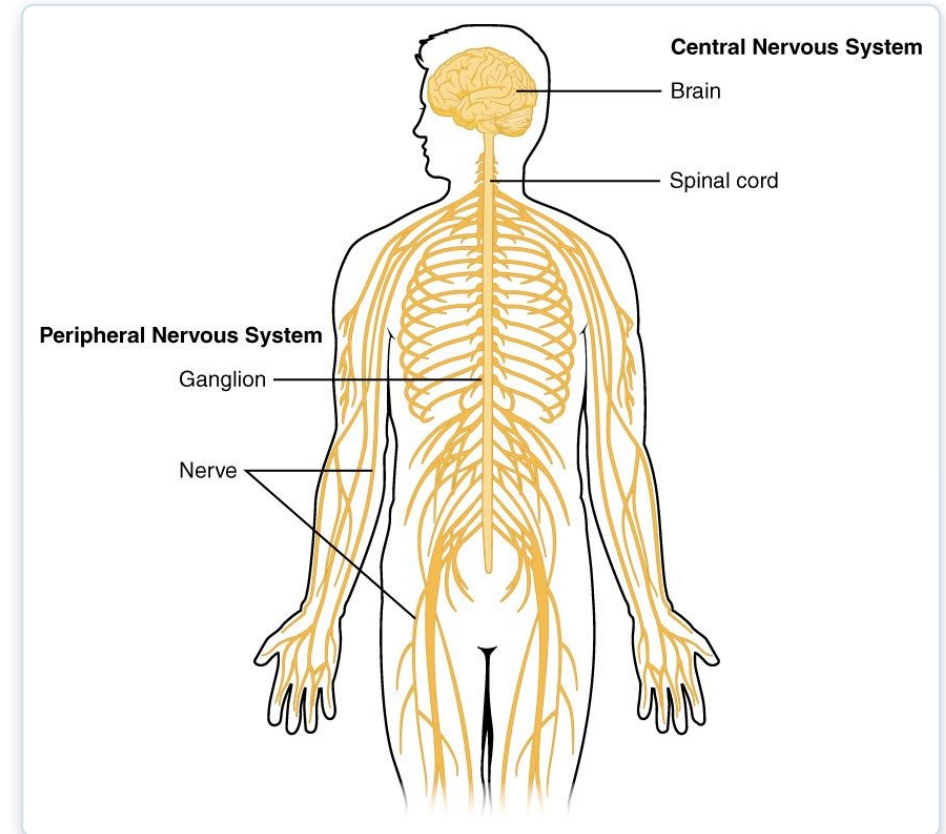
Neurons, glia, and how nerve signals are generated and passed.



Learning Objectives

After studying this chapter, students will be able to:

- 1 Name the major divisions of the nervous system, both anatomical and functional
- 2 Describe the functional and structural differences between gray matter and white matter structures
- 3 Name the parts of the multipolar neuron in order of polarity
- 4 List the types of glial cells and assign each to the proper division of the nervous system, along with their function(s)
- 5 Distinguish the major functions of the nervous system: sensation, integration, and response
- 6 Describe the components of the membrane that establish the resting membrane potential



The nervous system: central and peripheral divisions.



12.1

SECTION

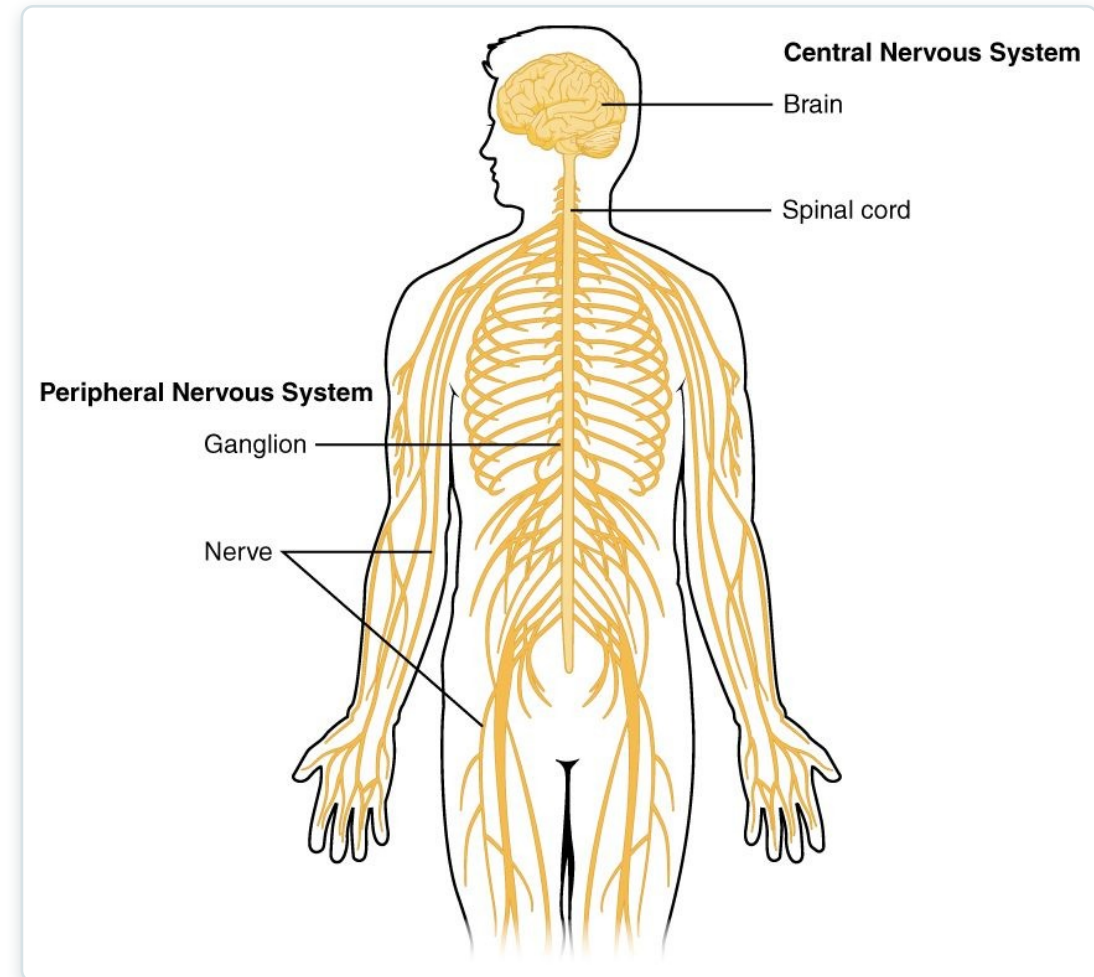
Basic Structure and Function of the Nervous System

The organization of the nervous system into central and peripheral divisions.

SECTION 12.1

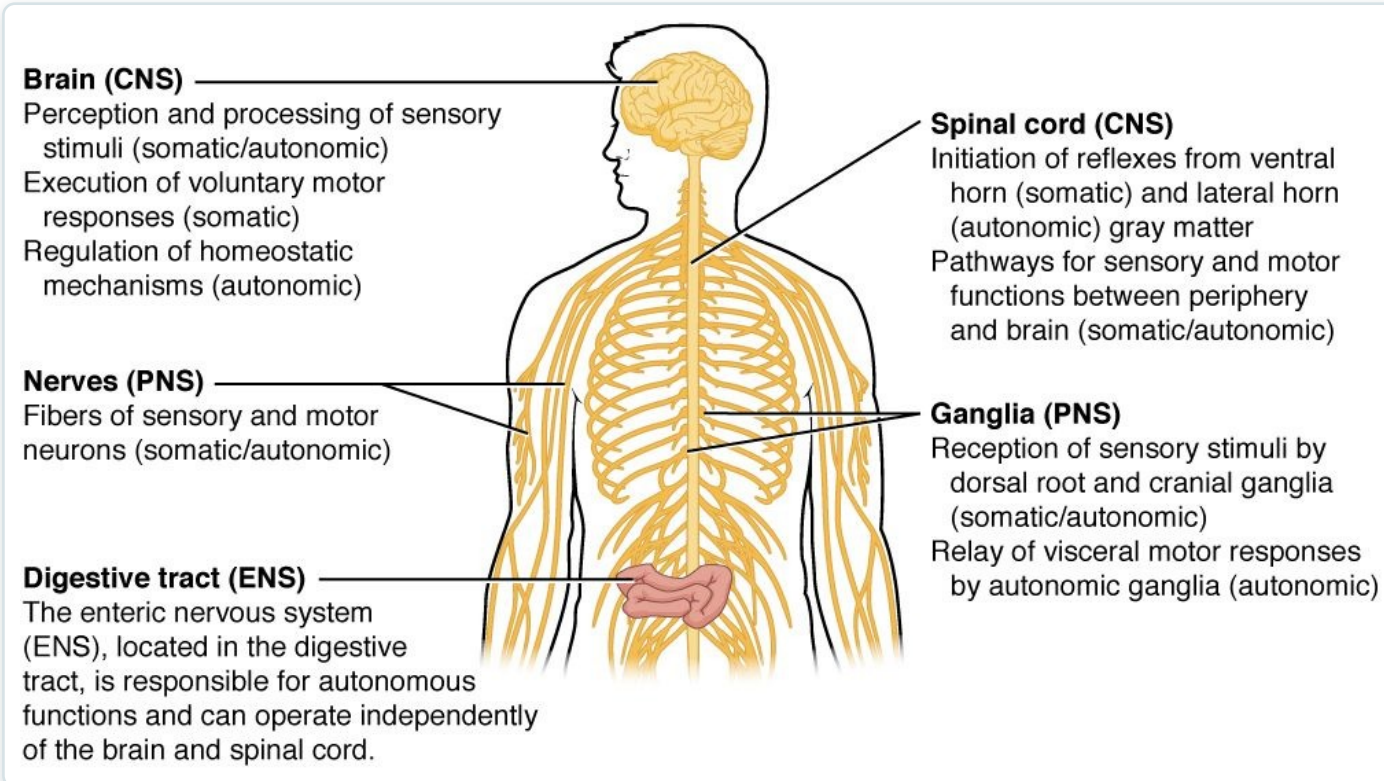
Central & Peripheral Nervous Systems

- Two main anatomical divisions
- CNS = brain and spinal cord
- PNS = nerves and ganglia
- CNS integrates, PNS connects



The central and peripheral nervous systems

Functional Divisions



The functional divisions of the nervous system

- Sensory (afferent) input to the CNS
- Motor (efferent) output from the CNS
- Somatic controls voluntary muscle
- Autonomic controls involuntary functions

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12.2

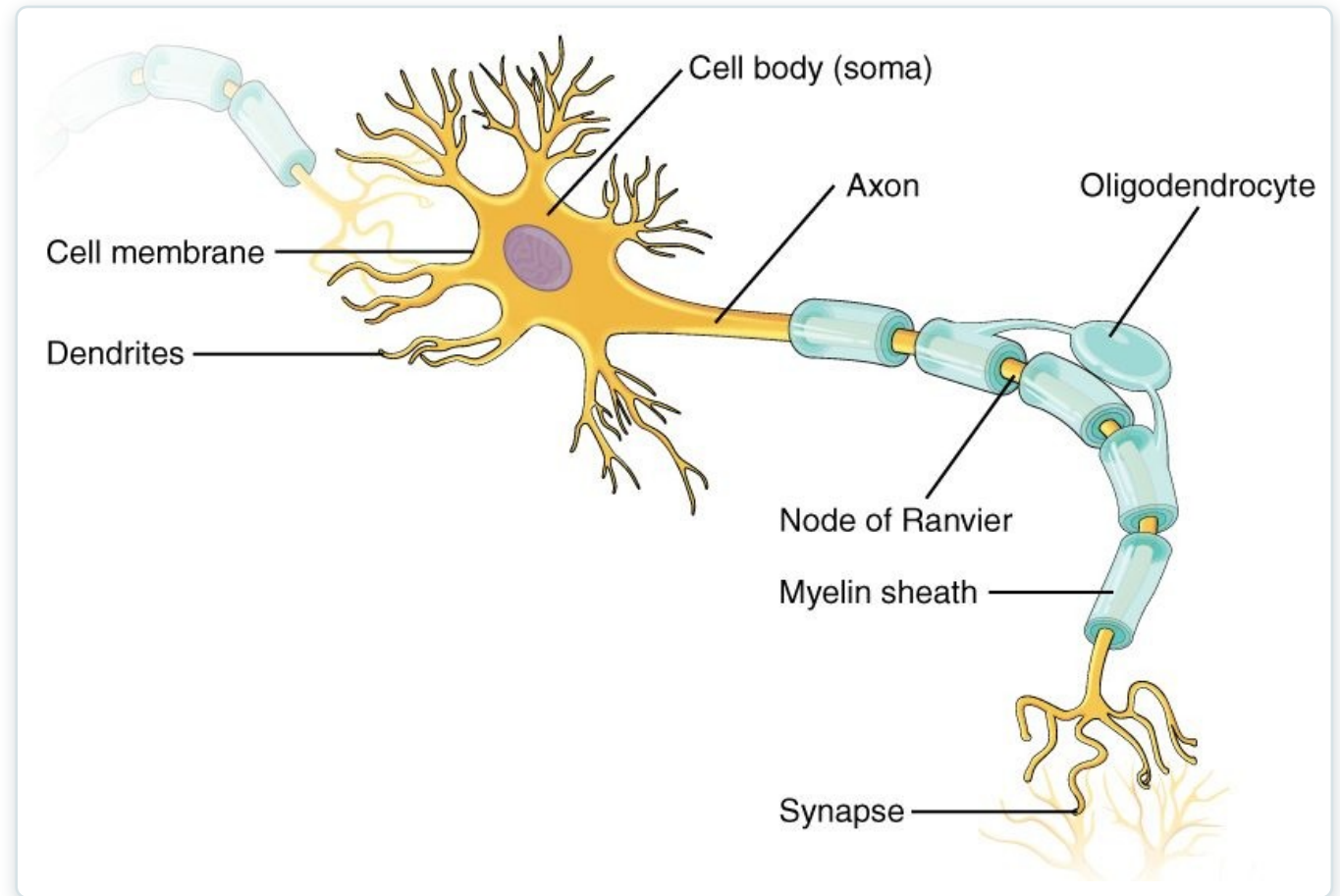
SECTION

Nervous Tissue

Neurons and the supporting glial cells that make up nervous tissue.

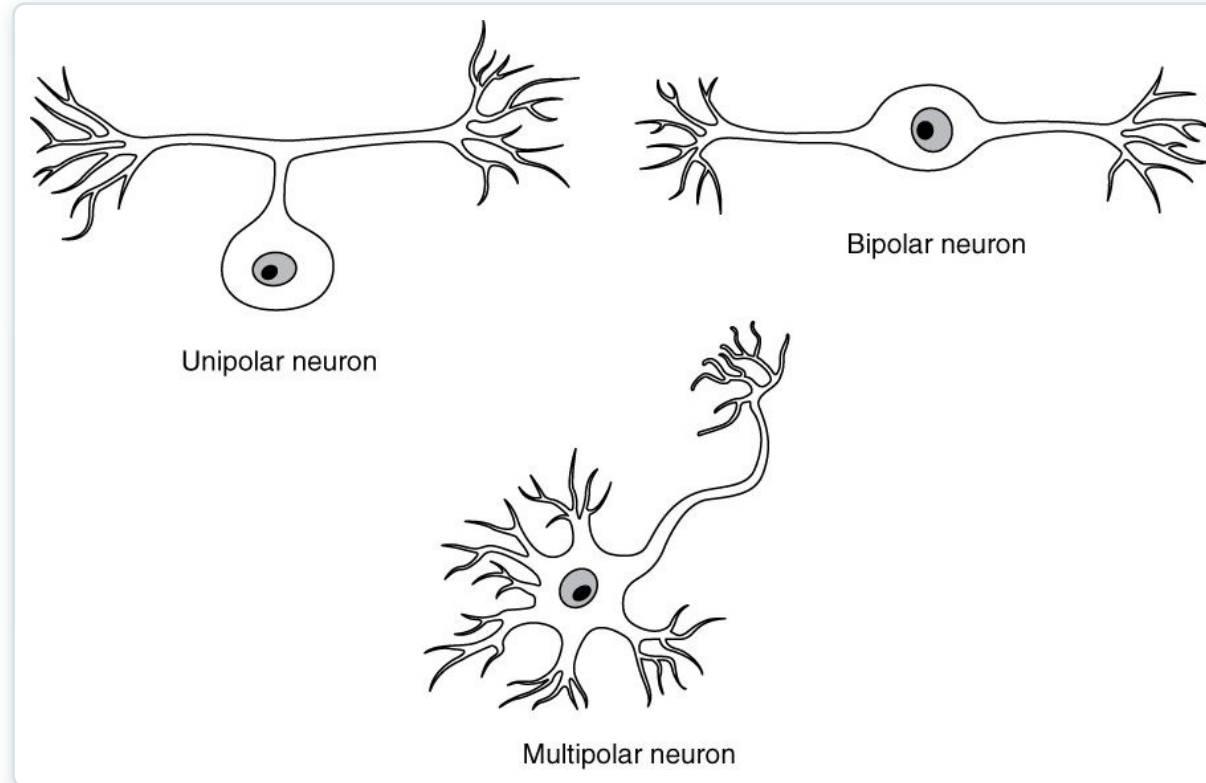
The Neuron

- The signaling cell of the system
- Dendrites receive incoming signals
- Cell body (soma) integrates them
- Axon carries the signal away



The structure of a neuron

Types of Neurons



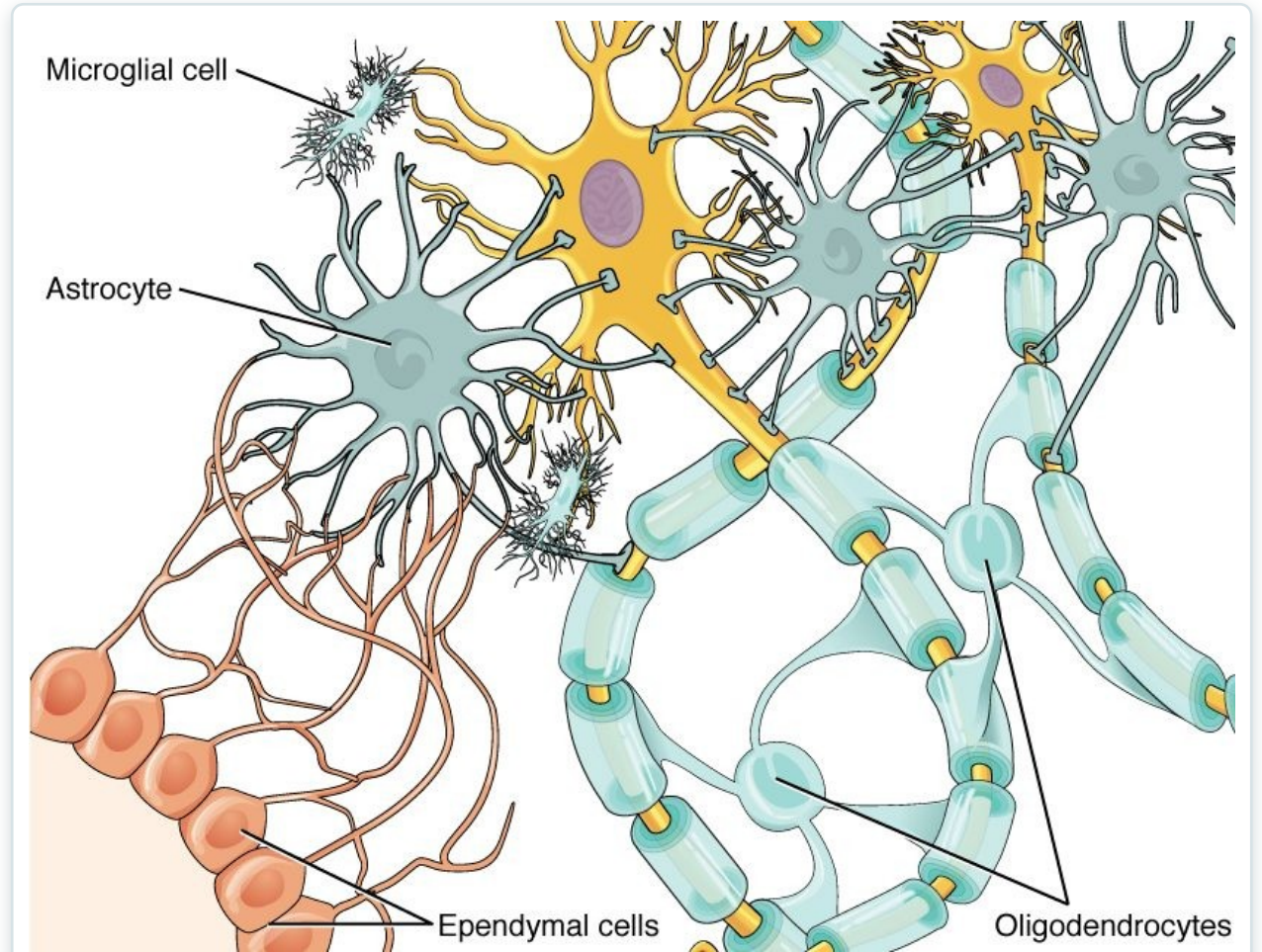
Structural classes of neurons

- Classified by shape (processes)
- Multipolar — most common (CNS)
- Bipolar — in special senses
- Unipolar — most sensory neurons

SECTION 12.2

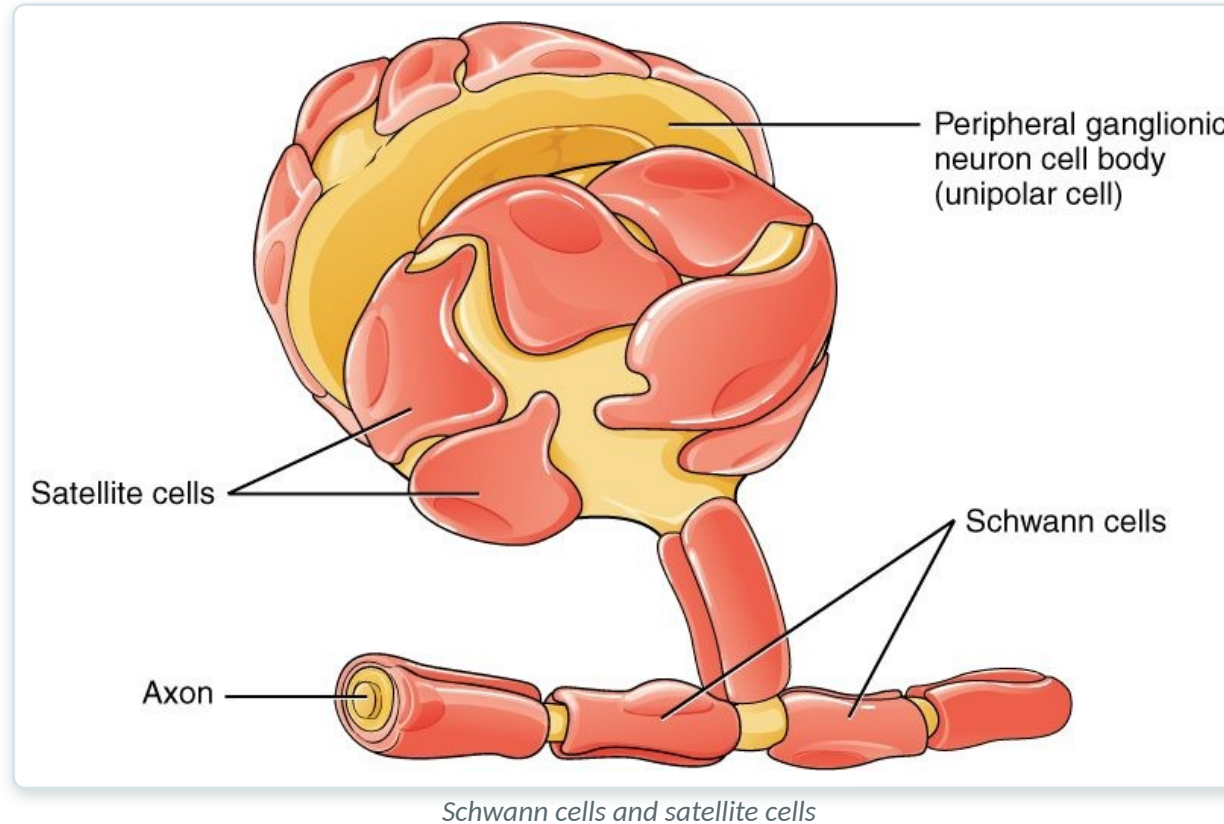
Glial Cells of the CNS

- Glia support and protect neurons
- Astrocytes maintain the environment
- Microglia are immune defenders
- Oligodendrocytes myelinate CNS axons



The glial cells of the CNS

Glial Cells of the PNS



- Schwann cells myelinate PNS axons
- One Schwann cell per segment
- Satellite cells support cell bodies
- Aid repair of peripheral nerves



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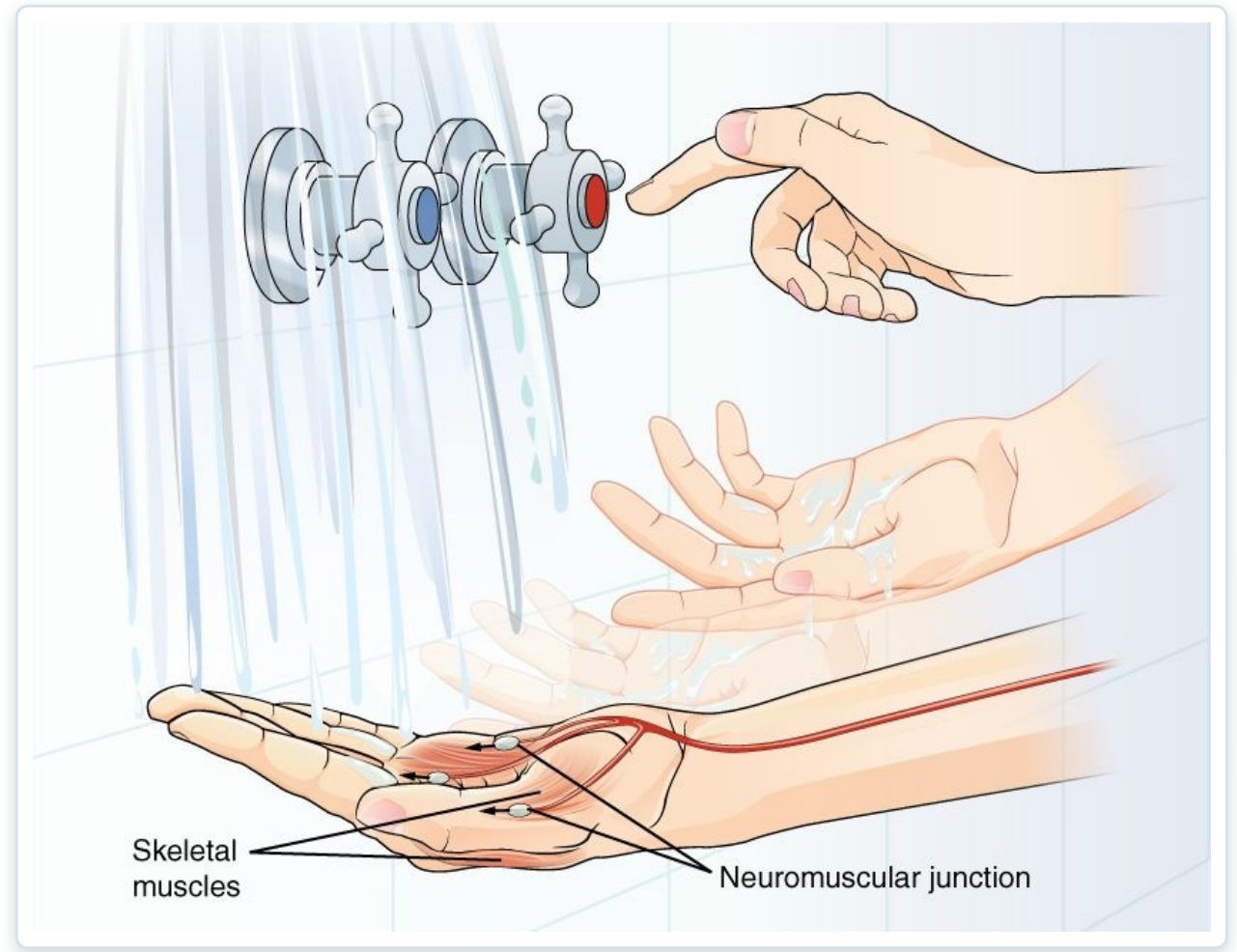
SECTION

The Function of Nervous Tissue

How neurons detect stimuli and carry signals — the reflex arc.

The Reflex Arc

- The simplest neural pathway
- Receptor → sensory neuron → CNS
- Motor neuron → effector response
- Produces fast, automatic reflexes



A withdrawal reflex arc

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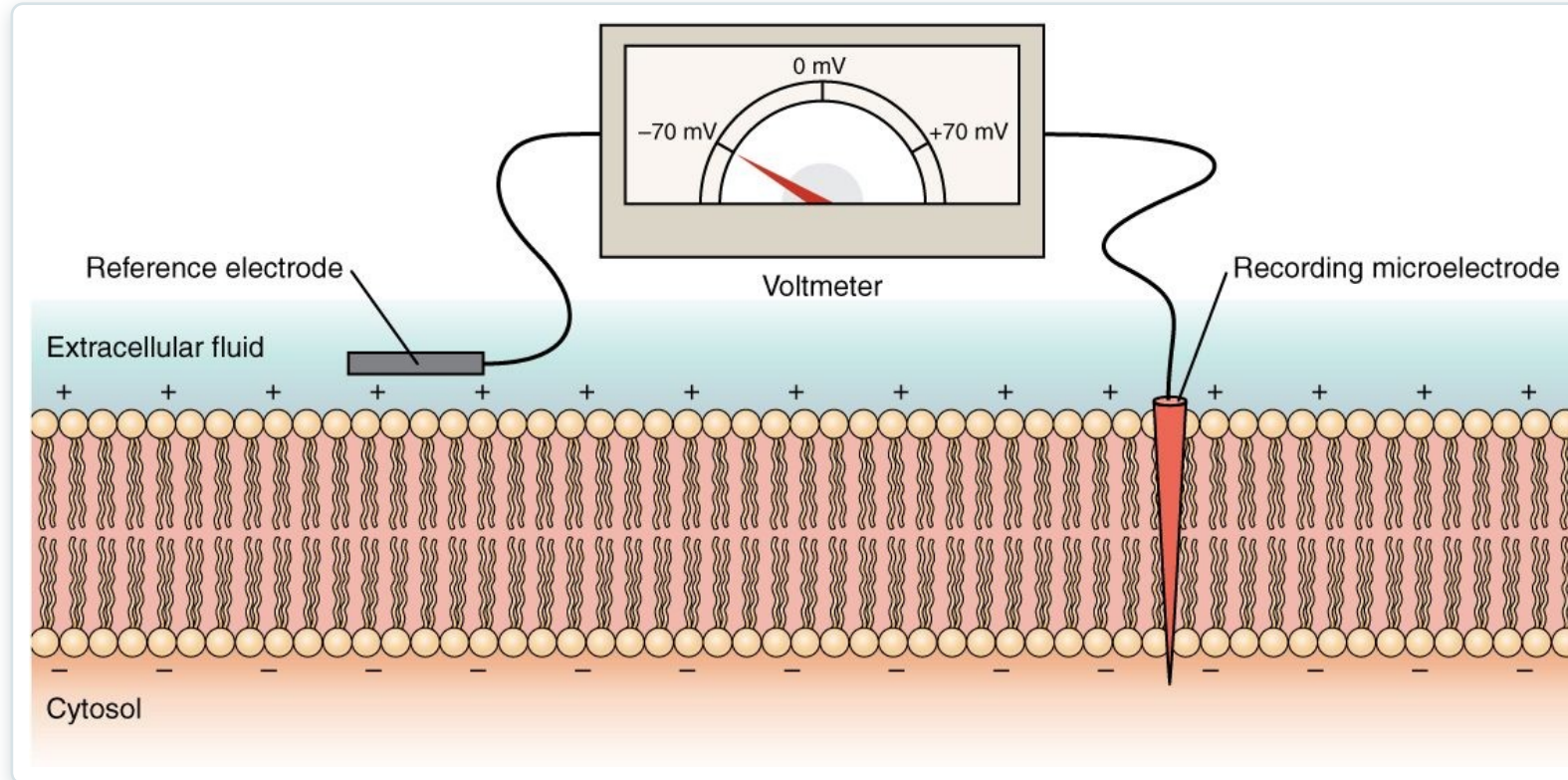
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SECTION

The Action Potential

The electrical signal — how neurons generate and conduct impulses.

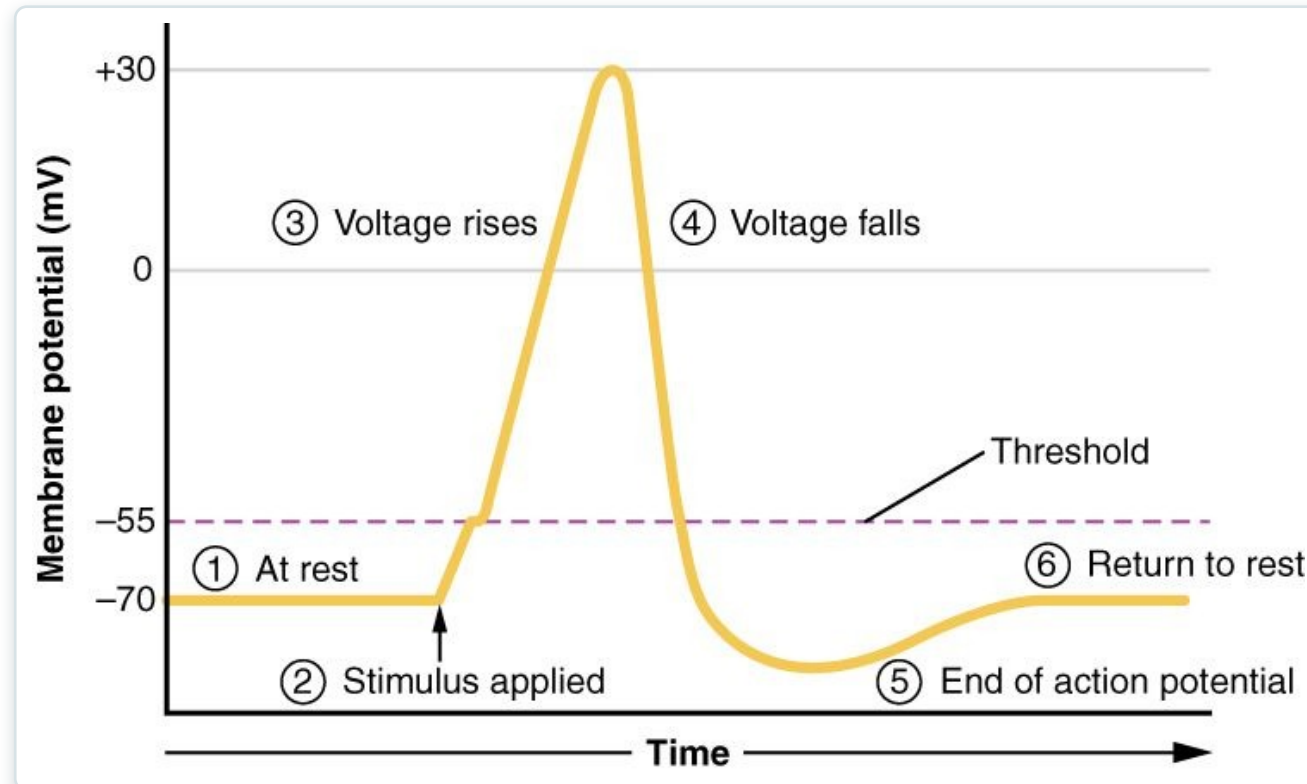
The Resting Membrane Potential



Measuring the resting membrane potential

- Neurons are electrically polarized
- Inside is negative (about -70 mV)
- Maintained by pumps and channels
- Ready to fire when stimulated

The Action Potential



The phases of an action potential

- An all-or-nothing electrical impulse
- Threshold triggers depolarization
- Sodium in, then potassium out
- Self-propagates along the axon



12.5

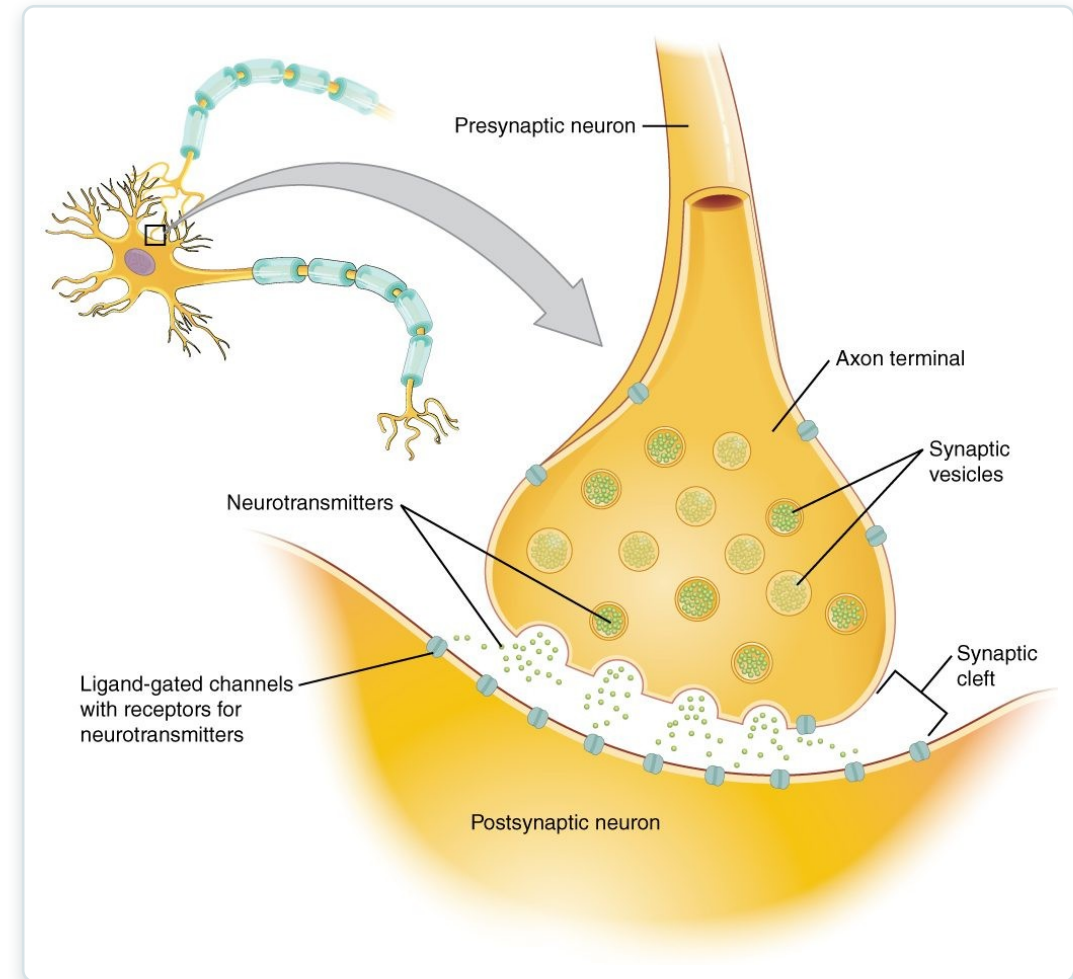
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Communication Between Neurons

How neurons pass signals to other cells — the synapse.

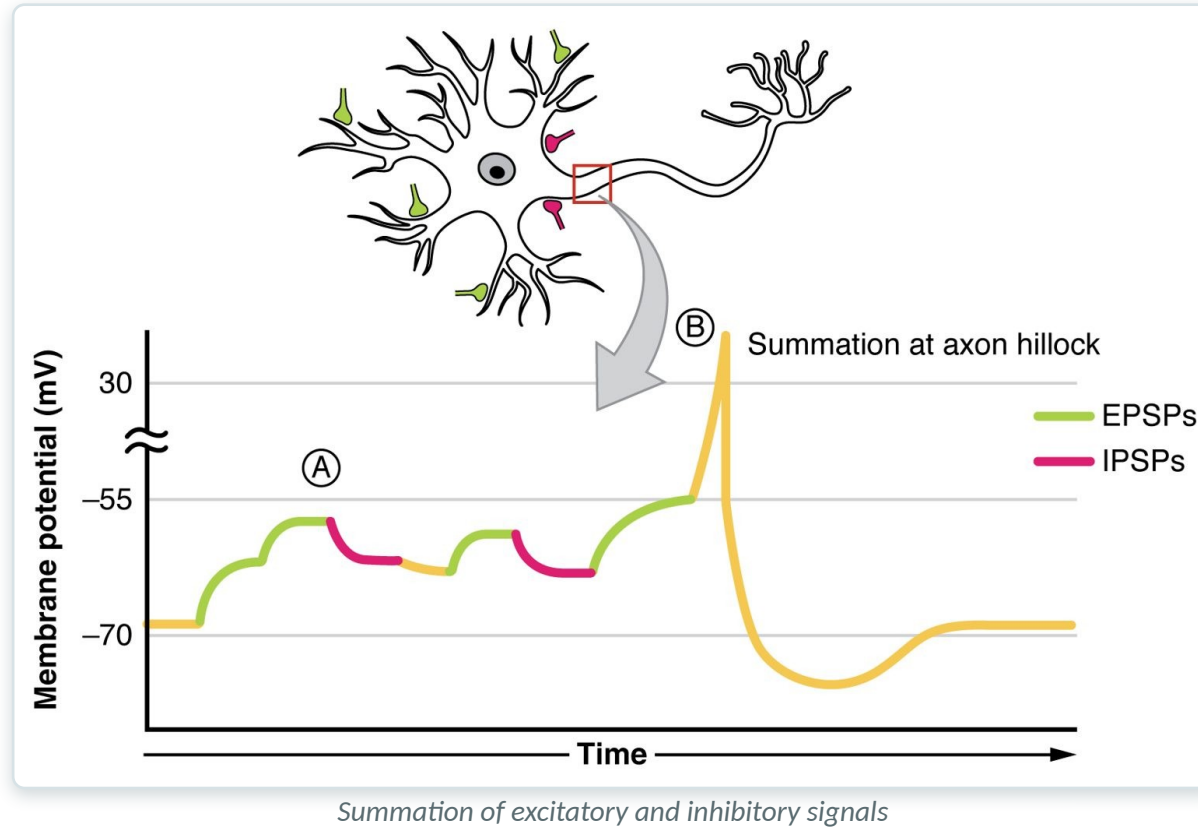
The Synapse

- Neurons communicate at synapses
- Impulse triggers neurotransmitter release
- Chemical crosses the synaptic cleft
- Binds receptors on the next cell



Structure of a chemical synapse

Synaptic Integration



- Inputs can excite or inhibit
- EPSPs push toward firing
- IPSPs push away from firing
- Summation decides the outcome

Key Takeaways

- The nervous system divides into the CNS (brain and spinal cord) and the PNS (nerves and ganglia).
- Neurons carry signals; glial cells support, protect, and myelinate them.
- The reflex arc is the simplest functional pathway, producing fast automatic responses.
- Neurons signal electrically through resting and action potentials.
- Neurons communicate at synapses, integrating excitatory and inhibitory input.

CLINICAL CONNECTIONS

Why Nervous Tissue Matters in Practical Nursing

Multiple Sclerosis

Loss of CNS myelin slows or blocks nerve signals.

Neuropathy

Peripheral nerve damage causes numbness, tingling, and weakness.

Reflex Testing

Reflexes are a quick window into nervous system function.

Anesthetics

Local anesthetics block sodium channels to stop pain signals.

Synaptic Drugs

Many medications act on neurotransmitters at synapses.

Nerve Injury

Peripheral nerves can regenerate; CNS neurons largely cannot.